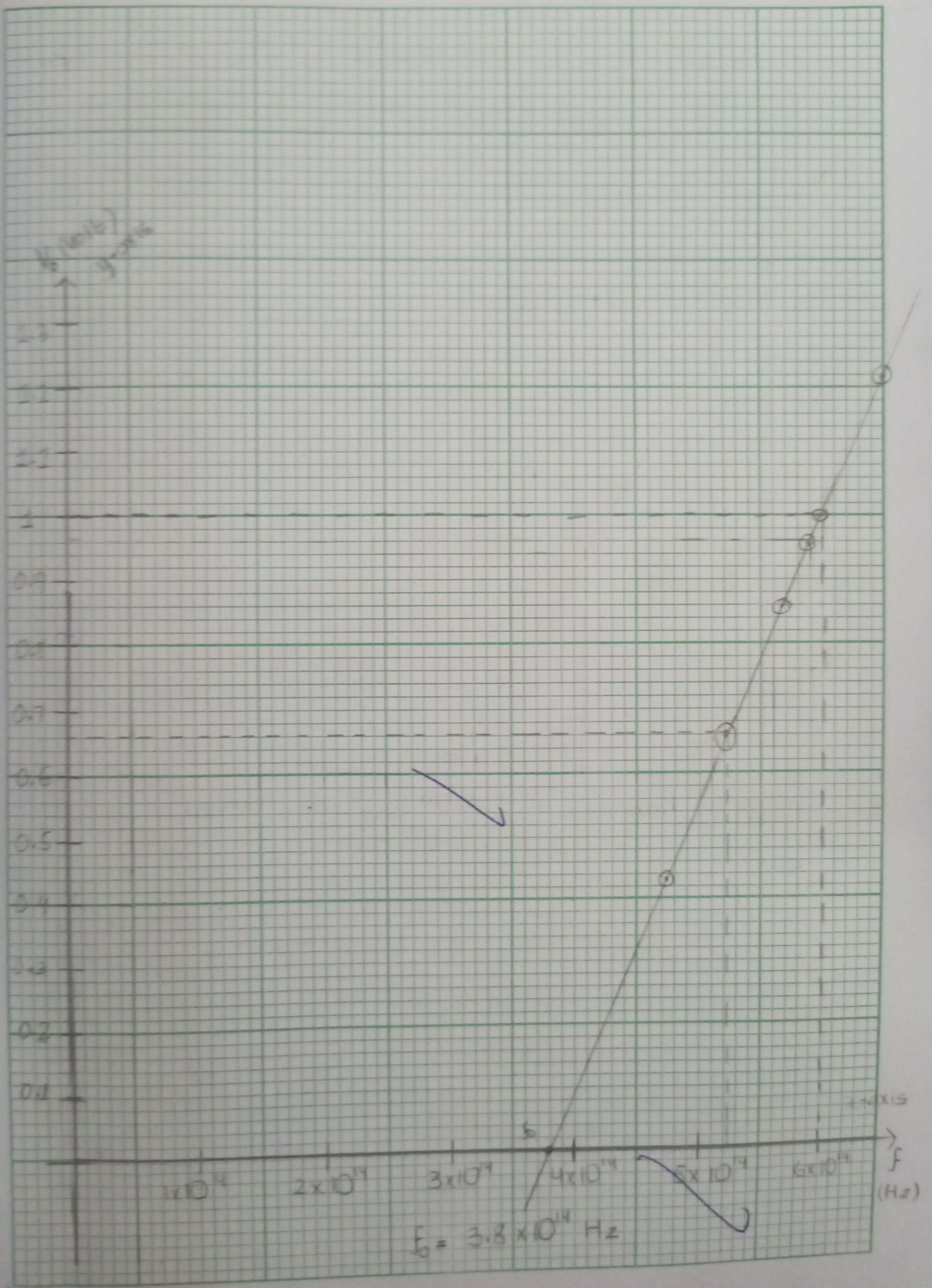


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On x-axis 10 small boxes
= 1×10^{14} Hz
On y-axis 5 small boxes
= 0.1 volt



EXPERIMENT # 02

OBJECT:

To find the ionization potential of mercury using a gas filled diode.

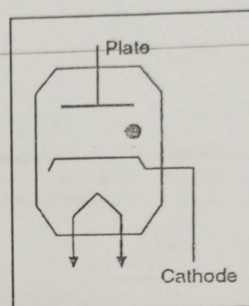
APPARATUS:

A gas filled (mercury vapors) diode, a D.C. power supply, a voltmeter, micro-ammeter, a potentiometer.

THEORY:

Ionization potential: The least energy required to remove the most loosely bound electron from an isolated free neutral atom is known as the ionization potential of the atom.

Ionization potential of mercury: The ionization potential of mercury can be determined by introducing mercury vapor at a low pressure of 10 mm to 50 mm of mercury column in an evacuated tube fitted with a cathode and an anode. A mercury vapor filled gas diode is the most suitable for the purpose. The cathode of the gas diode may be directly or indirectly heated type. A hot cathode gas filled diode is known as a *phanotron* [Incidentally a gas filled triode is known as a *Thyratron*] A gas filled diode is symbolically represented as shown in Fig. The dot in the tube shows the presence of the gas (or vapor).



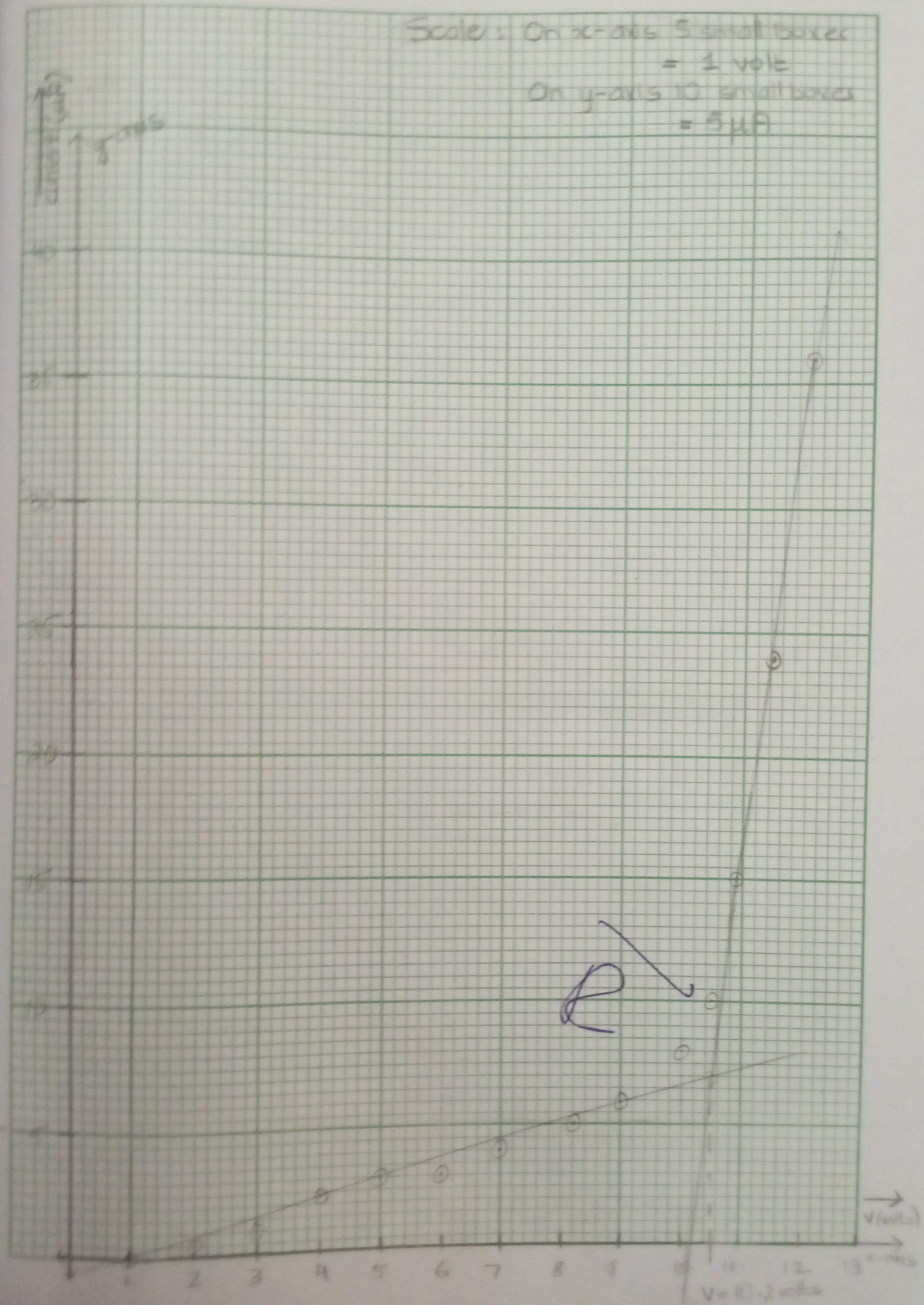
When the anode or plate of the gas filled diode is at a positive potential with respect to the cathode, electrons move across the tube from the cathode to the anode. This electronic current depends upon two factors:

- (i) The number of electrons emitted per unit area from the cathode and its temperature.
- (ii) The effect of space-charge – the negatively charged region containing the electron cloud due to the accumulation of electrons emitted by the cathode.

As the plate potential is increased the plate current slowly increases. But when the plate potential is increased beyond a particular value the plate current increases much more rapidly than it does below that critical value. This is because when the plate potential approaches this critical value the electrons arriving at the anode gain enough energy to knock out the electrons from the atoms of the gas close to the anode. These electrons are also attracted by the anode causing an increase in plate current and the positive ions neutralize some of the space charge, which further helps to increase the kinetic energy of the thermo-electrons. This potential is equal to the ionization potential of the gas (or vapor) and for this value of plate potential there is a marked increase in plate current.

If a graph is plotted between plate potential and plate current (for a constant value of filament current) the plate current at first increases slowly for a given increase in plate voltage and when the plate potential is equal to or greater than the ionization

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EXPERIMENT # 03

OBJECT:

To determine the velocity of wave propagation in stretched string by using sonometer.

APPARATUS:

WA-9611 Sonometer, WA-9613 Driver/Detector Coils, Function Generator, Dual trace oscilloscope, Mass and mass hanger.

THEORY:

When two exactly same waves (i.e. same amplitude, frequency and time period) travel in opposite direction with same velocity superimpose on one another the resultant wave obtained is called *standing* or *stationary* wave.

Hence, we can say that the wave length of standing wave in a vibrating string may be a function of the tension, liner mass density of string, frequency and the velocity of the wave.

If λ is the wavelength of standing wave and v is their velocity, the frequency of the vibration f can be determined by the relation.

The velocity of transverse wave in a string is $v = f \lambda$ _____ (i)

$$v = \sqrt{\frac{T}{\mu}} \text{ _____ (ii)}$$

T is the tension applied to the string and μ is the liner mass density. Substituting the above value of v from (i) to (ii)

$$f \lambda = \sqrt{\frac{T}{\mu}}$$

$$f = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$$

whereas the distance l between successive nodes on string give the wavelength of string waves. When string is vibrating in one loop we have

$$l = \lambda/2$$

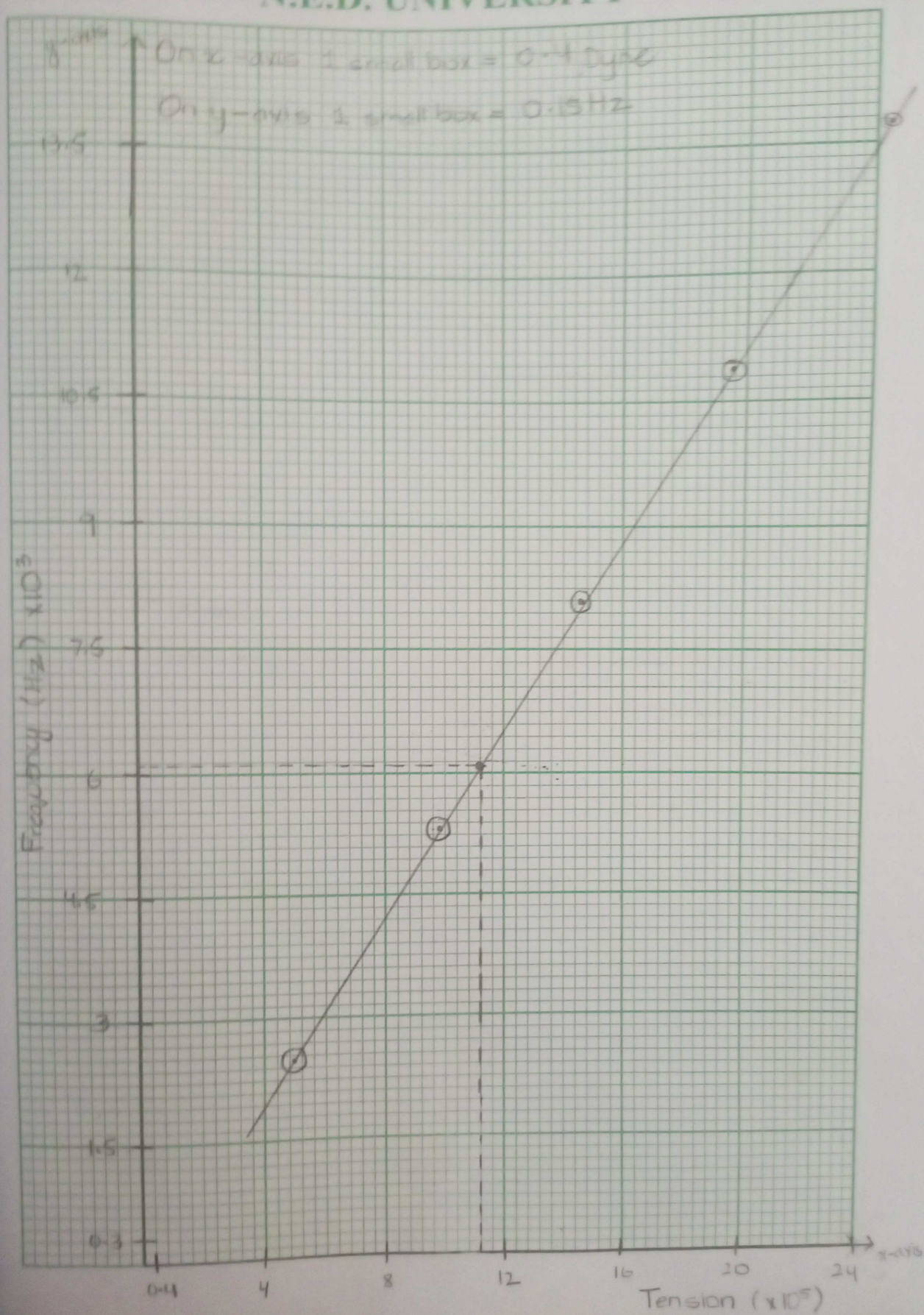
$$\lambda = 2l$$

and

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

The above equation gives the frequency of transverse wave in a string.

$$\mu = \frac{1}{4L^2} \frac{T}{f^2}$$



EXPERIMENT # 04

OBJECT:

To determine the horizontal component of Earth Magnetic field strength " H_{ex} " by Tangent Galvanometer.

APPARATUS:

Tangent galvanometer, coil, rheostat, reversing key, connecting wires and ammeter.

THEORY:

Two magnetic fields can be compared by means of tangent galvanometer, which consist of a magnet pivoted on a vertical axis and carrying a light pointer which can move over a circular scale. Normally one of the field is the earth horizontal component and other field is arranged to be at right angle to this. The pivoted magnet sets itself along the resultant of the two fields at an angle θ to its direction when it is in the earth magnetic field alone. If H_e is magnetizing force of the earth horizontal component and H is the magnetizing force of the other field (Coil).

$$H = H_{ex} \tan \theta \text{-----(1)}$$

The magnetizing force at the center of a circular coil of known radius and known number of turns, when a measured current is passing through the coil, can be calculated. This can be compared by means of tangent galvanometer, with the horizontal component of the earth's magnetizing force, enabling the latter to be determined.

If n is the number of turns in the coil, r meter is its radius, I amp is the current, and H amp per meter the magnetizing force at the center of the coil.

$$H = \frac{nI}{2r} \text{ (amp per meter)}$$

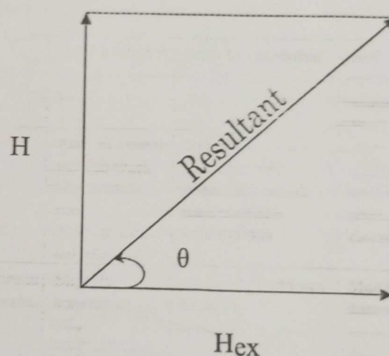
Using equation (1)

$$H = H_{ex} \tan \theta$$

$$H_{ex} = \frac{H}{\tan \theta}$$

$$H_{ex} = \frac{nI}{2r \tan \theta}$$

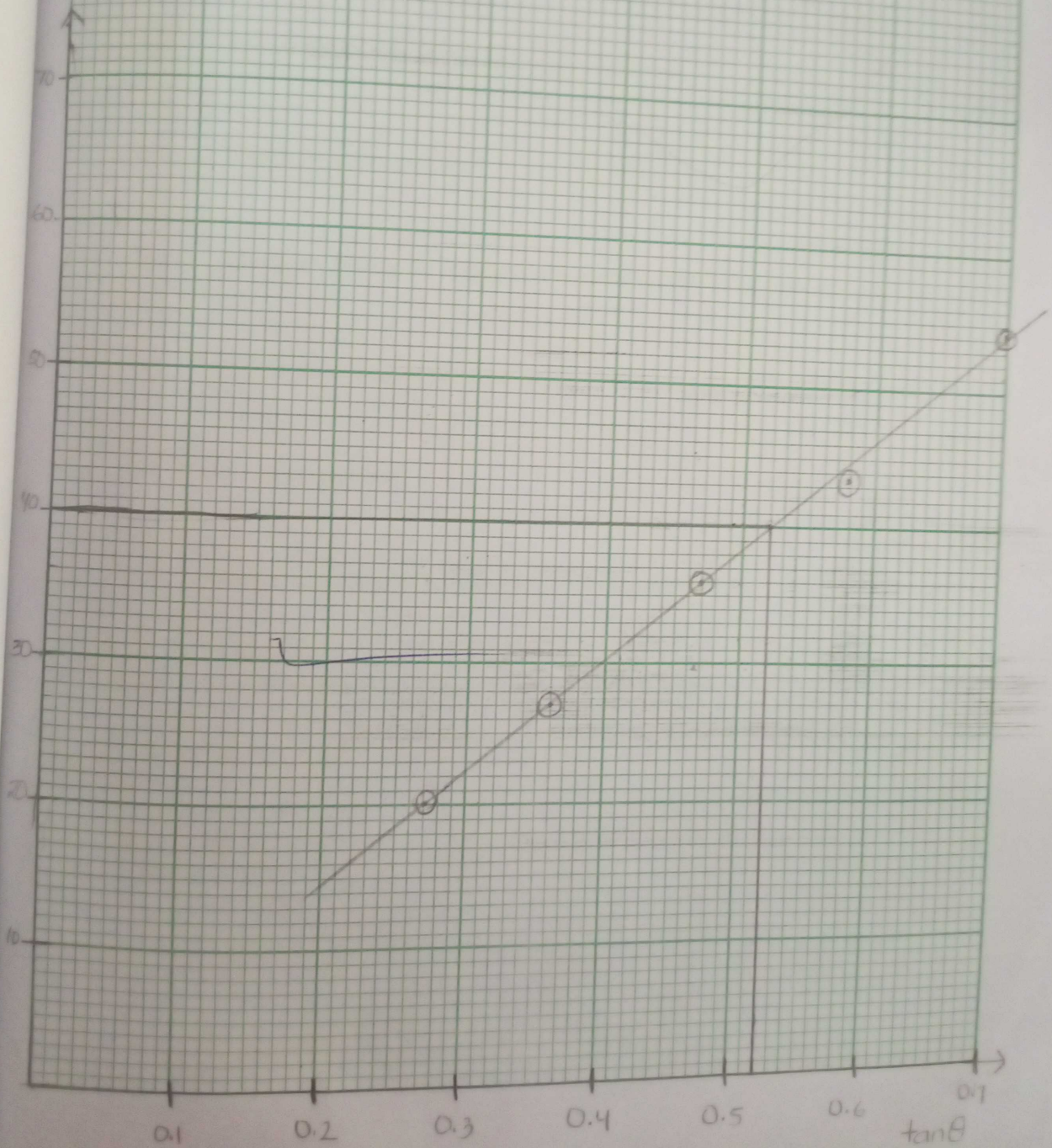
1 amp per meter = 0.01256 oersted



On x-axis 10 small boxes = 0.1°C

On y-axis 10 small boxes = $10\ \mu\text{f}$

Σ (μf)



EXPERIMENT # 05

OBJECT:

To study the characteristics of an acceptor circuit and determine unknown inductance.

APPARATUS:

Resistance, capacitors, inductor, a frequency generator/oscillator, an Oscilloscope, connecting wires, A.C. supply etc.

THEORY:

Acceptor Circuit:

When the resistor R, inductor L and capacitor C are connected in series with a source of emf E, the circuit is known as the RLC-series or series resonant or series tuned circuit as shown in figure.

The total opposition offered by the RLC circuit in flow of current is called impedance Z. it depends on frequency of AC signal applied and given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$ are inductive and capacitive reactance of inductor and capacitor respectively

In A.C. circuits the voltage and the current are usually not in phase. Across the inductor, the current lags behind the voltage by 90° , whereas across the capacitor, the current leads the voltage by 90° . But across the resistor the voltage and current both are in phase. At particular frequency the capacitive reactance X_C and the inductive reactance X_L become equal and cancel effect of each other and circuit behave as purely resistive circuit. Under this condition, the voltage and current are in phase, even though the circuit consists of L, C and R. This phenomenon is called resonance and the frequency is known as resonant frequency.

$$X_L = X_C$$

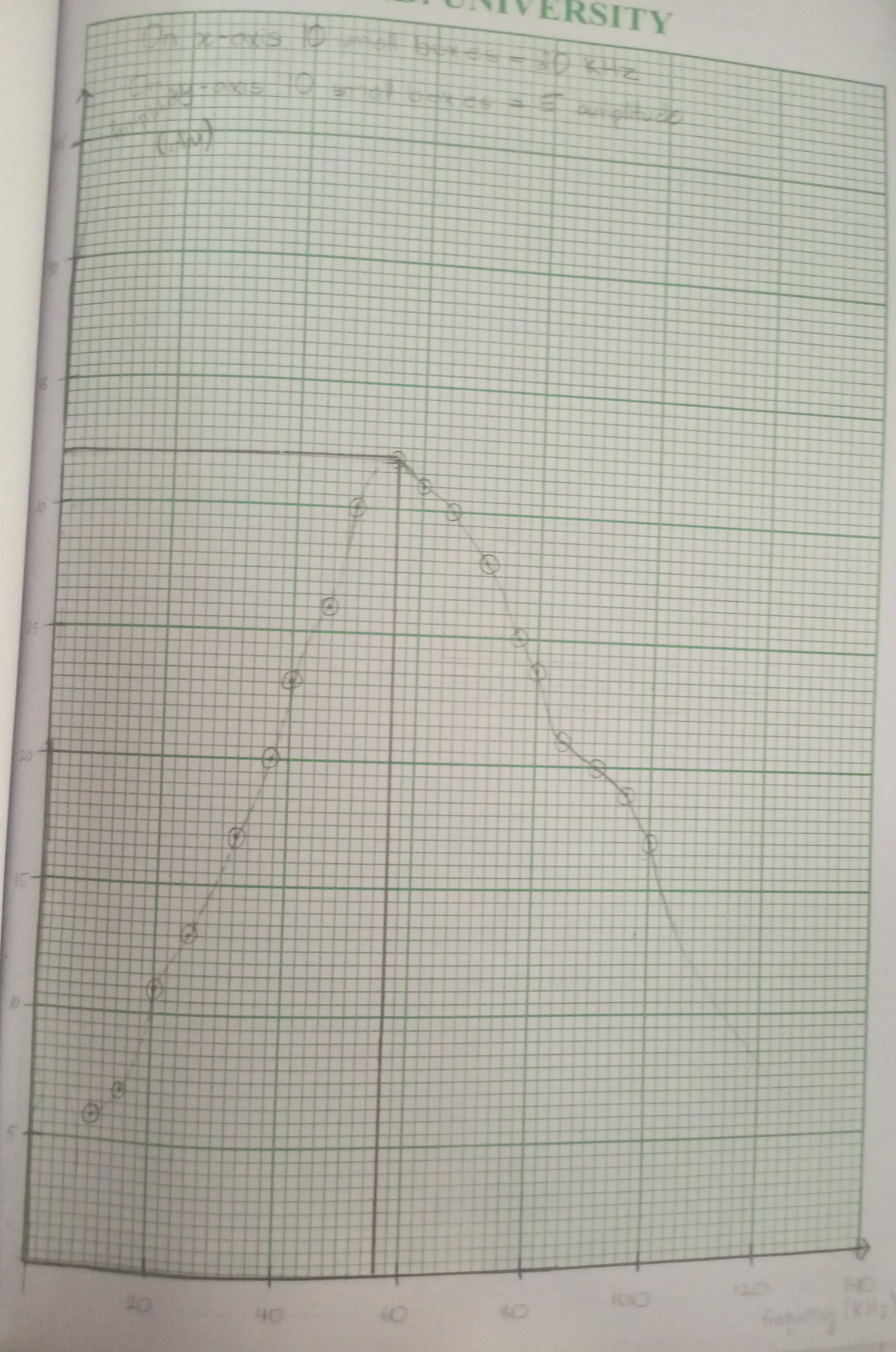
$$\omega L = \frac{1}{\omega C}$$

$$\omega = 2\pi f = \frac{1}{\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

This is an acceptor circuit, that means it allows maximum current to flow through it at this particular (resonant) frequency and at all other frequencies it allows less current.

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Equipment
 e Calibre
 Due date

Signed: _____

EXPERIMENT # 06

OBJECT:

To determine the unknown high resistance by Neon Flash lamp Apparatus.

APPARATUS:

Neon lamp, known and unknown high resistance, a capacitor of suitable value, stops watch and power peck etc.

THEORY:

Flashing and quenching of a neon bulb. A neon bulb is placed in parallel with a capacitor and connected to D.C. supply which can be continuously increased from 0 to 150 volt through a high resistance of about $1\text{M}\Omega$. The voltage is slowly increased to a value say V_1 when the lamp flashes and begins to glow. As soon as the neon lamp flashes, it becomes conducting and the capacitor begins to discharge through it. It continues to do so until the extinction (or quenching) potential V_2 is reached when the neon lamp ceases to glow and stops conducting. The capacitor then again begins to charge till the flashing potential V_1 is reached when again the lamp flashes and begins to glow. The process is repeated. During the time the capacitor is charging the neon lamp does not glow. In other words, the total time t between two consecutive flashes is equal to the time taken by the voltage first to fall from the flashing potential V_1 to quenching potential V_2 (discharge) and then to rise from V_2 to V_1 (charging). This flashing and quenching time can be determined by noting the time taken by the lamp to produce say 20 consecutive flashes and quenches. If t_1 is the time taken by the capacitor voltage to fall from V_1 to V_2 and t_2 is the time taken by the voltage to rise from V_2 to V_1 , then

$$V_2 \text{ for charging of capacitor, } V_2 = V_1 e^{-t_1/RC}$$

$$t_1 = -CR \log_e \frac{V_2}{V_1}$$

$$V_2 \text{ for discharging of capacitor, } V_2 = V_1 \left(1 - e^{-t_2/RC}\right)$$

$$t_2 = -CR \log_e \left(1 - \frac{V_2}{V_1}\right)$$

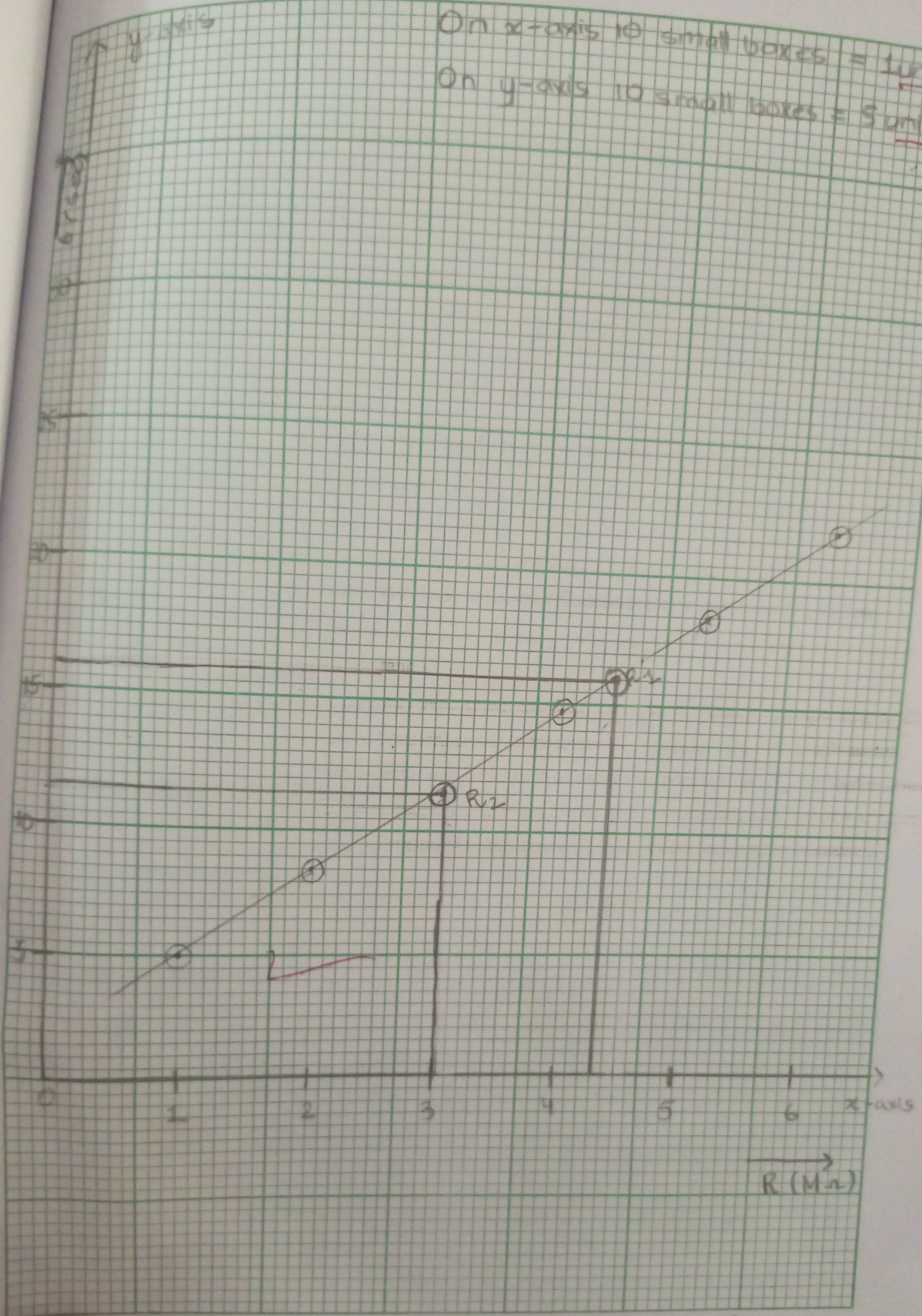
$$t = t_1 + t_2 = R \left[-C \log_e \frac{V_2}{V_1} - C \log_e \left(1 - \frac{V_2}{V_1}\right) \right]$$

As C , V_2 and V_1 have constant fixed values, the function within brackets is a constant, and denoted by k

$$\therefore t = kR$$

$$t \propto R$$

On x-axis 10 small boxes = 1 unit
 On y-axis 10 small boxes = 5 unit



$$R_1 = 4.4 - 1 = 3.4 \text{ Mn}$$

$$R_2 = 3 - 1 = 2 \text{ Mn}$$